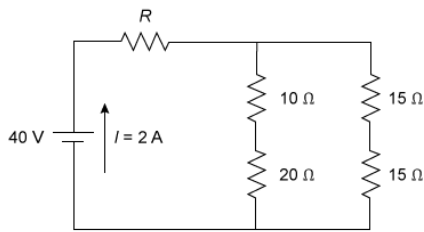


In the circuit shown a current of 2 A flows through the 40-V battery.



What is the value of the unknown resistance R ?

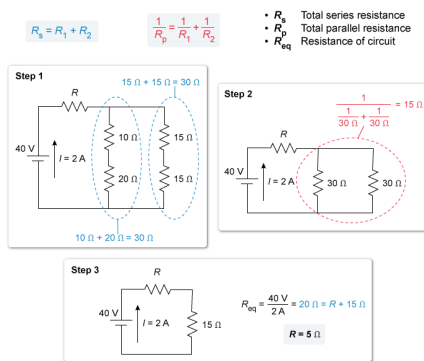
- ✓ ☒ A. 5 Ω [47%]
☐ B. 15 Ω [9%]
☐ C. 20 Ω [37%]
☐ D. 30 Ω [5%]

Correct

47% answered correctly

Explanation:

Combining resistors in a circuit with series and parallel resistors



In an electric circuit, a voltage generates a **current** I , which is the flow of charge per unit time in a conductor (eg, wire, bulb, resistor). In addition, **Ohm's law** states that voltage V is equal to the product of resistance R and current I :

$$V = IR$$

In a circuit with multiple resistors in series and parallel, a total equivalent resistance R_{eq} for the entire circuit must be calculated. The total resistance of resistors in **series** R_s and in **parallel** R_p can be combined as:

$$R_s = R_1 + R_2 + \dots + R_n$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

In this question, the equivalent resistance of the entire circuit is found using Ohm's law:

$$R_{eq} = \frac{V}{I} = \frac{40\ V}{2\ A} = 20\ \Omega$$

In this circuit, each of the parallel branches of resistors contains two resistors in series, so each branch's equivalent resistance is the sum of the resistors in each branch:

$$R_{s1} = 10\ \Omega + 20\ \Omega = 30\ \Omega$$

$$R_{s2} = 15\ \Omega + 15\ \Omega = 30\ \Omega$$

Next, the equivalent resistance R_p of the parallel combination of R_{s1} and R_{s2} is:

$$\frac{1}{R_p} = \frac{1}{30\ \Omega} + \frac{1}{30\ \Omega} = \frac{2}{30\ \Omega}$$

Solving for R_p :

$$R_p = \frac{30 \ \Omega}{2} = 15 \ \Omega$$

Since the unknown resistance R is in series with R_p , the equivalent resistance of the entire circuit is:

$$R_{eq} = R + R_p = R + 15 \ \Omega$$

Substituting the value of R_{eq} and solving for R :

$$20 \ \Omega = R + 15 \ \Omega$$

$$R = 5 \ \Omega$$

Therefore, the value of the unknown resistance R is $5 \ \Omega$.

(Choice B) $15 \ \Omega$ is the equivalent resistance of the parallel branches, not the value of only R .

(Choice C) $20 \ \Omega$ is the equivalent resistance of the entire circuit, not the value of only R .

(Choice D) $30 \ \Omega$ is the resistance in each of the parallel branches.

Educational objective:

The total resistance in an electric circuit increases as resistors are added in series and decreases as resistors are added in parallel. Ohm's law relates resistance to the voltage and current.

Time spent: 0 sec

Subject:
Physics

QID: 404051

System:
Electrostatics and Circuits